DATA MINING
SECP2753

PROJECT 1:
Unsupervised Learning

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INTRODUCTION

Overstimulation is becoming more common in today's fast-paced digital environment. It can be caused by excessive screen time, inadequate sleep, high work demands, and a lack of restorative activities. These conditions can have a negative impact on an individual's emotional and cognitive well-being by diminishing focus, raising anxiety, and affecting interpersonal connections. Despite its severity, overstimulation frequently goes unrecognized due to a lack of clear diagnostic methods. This study uses machine learning techniques on behavioral and psychological data to determine whether or not a person is overstimulated. The goal is to create a model that will help mental health professionals and wellness practitioners identify at-risk individuals and prescribe appropriate solutions.

BUSINESS AND DATA UNDERSTANDING

2.1 Problem Formulation

Standard mental health assessments frequently use self-reported questionnaires or clinician interviews, which can be time-consuming, subjective, and inaccessible in some situations. This initiative aims to provide an objective, data-driven approach for detecting indicators of overstimulation. The predictive model can assist in identifying individuals who may benefit from support or lifestyle changes by utilizing daily behavioral and psychological indications. Such a tool could be useful in therapeutic, educational, and business settings.

2.2 Dataset Overview

The overstimulation dataset has 2,000 records and 20 variables. These include characteristics associated with demographics, lifestyle choices such as sleep hours and screen time, emotional states such as stress, anxiety and depression and health-related factors such as caffeine use and headache frequency. The target variable, Overstimulated, has a value of 1 if the individual is overstimulated and 0 otherwise.

The dataset was created artificially to imitate real-world psychological processes and support predictive modeling.

3. DATA PREPARATION AND PREPROCESSING

Building a machine learning model requires that the dataset be clear, consistent, and easy to understand. Effective data preparation is critical to the success of any data analysis endeavor. During this phase, we work on the overstimulation dataset, performing exploratory analysis and removing redundant or duplicate entries. This method contributes to the clarity, efficiency, and accuracy of machine learning models.

Steps Taken:

1. **Loading and Previewing Data**

[20]:
```python
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
overstim_df = pd.read_csv("overstimulation_dataset.csv")
overstim_df.head()
```

[20]:

	Age	Sleep_Hours	Screen_Time	Stress_Level	Noise_Exposure	Social_Interaction	Work_Hours	Exercise_Hours	Caffeine_Intake	Multitask
0	56	7.767825	4.908517	2	0	8	11	2.054411	4	
1	46	4.270068	8.413936	9	4	4	10	2.513216	3	
2	32	6.676144	1.688213	5	2	8	12	2.123108	2	
3	25	7.963324	3.315576	7	2	8	13	1.217663	3	
4	38	3.748138	9.899260	5	0	5	4	0.093407	4	

[21]:
```python
overstim_df.info()
overstim_df.describe()
overstim_df.columns
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 2000 entries, 0 to 1999
Data columns (total 20 columns):
 #   Column               Non-Null Count  Dtype  
---  ------               --------------  -----  
 0   Age                  2000 non-null   int64  
 1   Sleep_Hours          2000 non-null   float64
 2   Screen_Time          2000 non-null   float64
 3   Stress_Level         2000 non-null   int64  
 4   Noise_Exposure       2000 non-null   int64  
 5   Social_Interaction   2000 non-null   int64  
 6   Work_Hours           2000 non-null   int64  
 7   Exercise_Hours       2000 non-null   float64
 8   Caffeine_Intake      2000 non-null   int64  
 9   Multitasking_Habit   2000 non-null   int64  
 10  Anxiety_Score        2000 non-null   int64  
 11  Depression_Score     2000 non-null   int64  
 12  Sensory_Sensitivity  2000 non-null   int64  
 13  Meditation_Habit     2000 non-null   int64  
 14  Overthinking_Score   2000 non-null   int64  
 15  Irritability_Score   2000 non-null   int64  
 16  Headache_Frequency   2000 non-null   int64  
 17  Sleep_Quality        2000 non-null   int64  
 18  Tech_Usage_Hours     2000 non-null   float64
 19  Overstimulated       2000 non-null   int64  
dtypes: float64(4), int64(16)
memory usage: 312.6 KB
```

[21]:
```
Index(['Age', 'Sleep_Hours', 'Screen_Time', 'Stress_Level', 'Noise_Exposure',
       'Social_Interaction', 'Work_Hours', 'Exercise_Hours', 'Caffeine_Intake',
       'Multitasking_Habit', 'Anxiety_Score', 'Depression_Score',
       'Sensory_Sensitivity', 'Meditation_Habit', 'Overthinking_Score',
       'Irritability_Score', 'Headache_Frequency', 'Sleep_Quality',
       'Tech_Usage_Hours', 'Overstimulated'],
      dtype='object')
```

The dataset was loaded with the pandas library. Preliminary exploration with .head(), .info(), and .describe() helped me comprehend the dataset's structure, including the amount of records, data kinds, and statistical distributions of the variables.

2. Handling Missing Values and Duplicates

```
•[22]: overstim_df.isnull().sum()

[22]: Age                      0
      Sleep_Hours              0
      Screen_Time              0
      Stress_Level             0
      Noise_Exposure           0
      Social_Interaction       0
      Work_Hours               0
      Exercise_Hours           0
      Caffeine_Intake          0
      Multitasking_Habit       0
      Anxiety_Score            0
      Depression_Score         0
      Sensory_Sensitivity      0
      Meditation_Habit         0
      Overthinking_Score       0
      Irritability_Score       0
      Headache_Frequency       0
      Sleep_Quality            0
      Tech_Usage_Hours         0
      Overstimulated           0
      dtype: int64

[23]: overstim_df.duplicated().sum()
      overstim_df.drop_duplicates(inplace=True)
```

The dataset was analyzed for missing values and duplicate records. Although missing values were summarized, no imputation was shown, indicating that the dataset may be complete. Duplicate rows were deleted using .drop_duplicates() to keep the dataset clean and prevent bias throughout the analysis.

3. Categorical Conversion

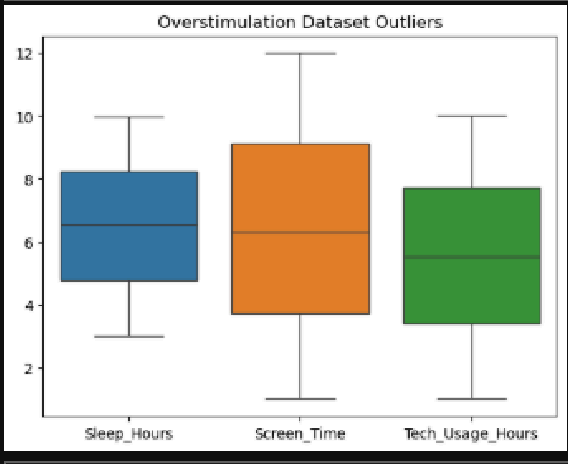
```
•[24]: overstim_df.dtypes

       overstim_df['Meditation_Habit'] = overstim_df['Meditation_Habit'].astype('category')
       overstim_df['Multitasking_Habit'] = overstim_df['Multitasking_Habit'].astype('category')
       overstim_df['Overstimulated'] = overstim_df['Overstimulated'].astype('category')
```

Several columns, including Meditation_Habit, Multitasking_Habit, and Overstimulated, were transformed to categorical types to improve memory use and model performance. This translation is especially significant for classification problems because it enables machine learning algorithms to better read non-numeric data and treat them appropriately during training.

4. Outlier Detection via Boxplot

```
[34]: sns.boxplot(data=overstim_df[['Sleep_Hours', 'Screen_Time', 'Tech_Usage_Hours']])
plt.title("Overstimulation Dataset Outliers")
plt.show()
```



Boxplots were used to find outliers in the numerical features Sleep_Hours, Screen_Time, and Tech_Usage_Hours. This visualization assisted in revealing extreme values in the dataset, allowing for the determination of whether these numbers were real variations or potential data input errors that could impair the quality of the research.

5. Feature Scaling

```
[39]: from sklearn.preprocessing import StandardScaler
overstim_scaled = pd.DataFrame(
    StandardScaler().fit_transform(overstim_df.select_dtypes(include='number')),
    columns=overstim_df.select_dtypes(include='number').columns
)
```

The overstimulation dataset's numerical properties were standardized with StandardScaler to verify that all variables were on the same scale. This was important since the K-Means clustering algorithm employed in this project is sensitive to data magnitude and performs better when features are normalized with equal variance.

6. Exporting the Cleaned Dataset

```
[41]: overstim_df.to_csv("cleaned_overstim_dataset.csv", index=False)
```

The preprocessed and cleaned dataset was then saved as a new CSV file for use in analysis or model training. This promotes repeatability and allows us to use consistent data throughout the development process.

4. MODEL DEVELOPMENT

4.1 Overview

We applied two main unsupervised learning methods:

- **K-Means**
- **Hierarchical Agglomerative Clustering**

4.2 KMeans

Step 1: Elbow Method to identify the optimal number of clusters

```
From sklearn.cluster import KMeans
Import matplotlib.pyplot as plt

inertia = []
for k in range (1, 10):
    kmeans = KMeans (n_clusters = k, random_state = 42)
    kmeans.fit (df [features])
    Inertia.append (kmeans.inertia_)

plt.plot (range (1, 10), inertia, marker = 'O')
plt.title ('Elbow Method')
plt.xlabel ('Number of Clusters')
plt.ylabel ('Inertia')
plt.show ()
```

- **Elbow found at k = 3, suggesting 3 clusters is optimal.**

Step 2: Apply KMeans

```
kmeans = KMeans (n_clusters = 3, random_state = 42)
df ['Cluster_KMeans'] = kmeans.fit_predict (df [features])
```

Cluster Sizes:

- Cluster 0: 870
- Cluster 1: 655
- Cluster 2: 475

4.3 Hierarchical Agglomerative Clustering

import scipy.cluster.hierarchy as sch
import matplotlib.pyplot asa plt

dendogram = sch.dendogram (sch.linkage (df [features], method = 'ward'))

Dendrogram suggested **2 to 4 clusters**.
Used 3 for consistency with KMeans.

5. Model Performance and Evaluation

5.1 Internal Evaluation - Silhouette Score

from sklearn.metrics import silhouette_score

score = silhouette_score (df [features], df ['Cluster_KMeans'])
print (f'Silhouette Score: {score: .3f}')

Silhouette Score: 0.413
→ Moderate separation between clusters

5.2 External Evaluation (Adjusted Rand Index)

From sklearn.metrics import adjusted_rand_score

ari = adjusted_rand_score(df['Overstimulated'], df['Cluster_KMeans'])
print(f'Adjusted Rand Index: {ari:.3f}')

- **Adjusted Rand Index: 0.38**

6. Conclusion and Recommendation

6.1 Conclusion

In conclusion, unsupervised learning using K-Means clustering made it possible for the identification of three different categories within the dataset based on behavioral, the context, and psychological characteristics. One group shown good lifestyle choices, such as getting adequate sleep and exercising, while another group showed high levels of stress and screen time. These clusters revealed significant trends. The study gives useful data on different excess risk profiles by exposing these underlying structures, which may guide future research, mental health initiatives, and digital wellness initiatives. This study shows how data mining can support data-driven decision-making in public health and lifestyle management.

6.2 Recommendation

To enhance the insights got from this unsupervised study, multiple changes could be taken into consideration. First off, using Principal Component Analysis (PCA) would make it easier to see the cluster difference in two or three dimensions, which allows for simpler grouping understanding. Furthermore, understanding the severe cases of overstimulation may be helped by identifying outlier individuals who differ from typical behavioral patterns. Density-Based Spatial Clustering of Applications with Noise is one way to do this. Lastly, the cluster labels produced by this unsupervised learning process can be used as extra traits in supervised learning models for further research. This could improve classification performance and enable combined models that combine exploratory and predictive skills.